

Algorithm Cheat Sheet — All Templates + Java Syntax

Conventions, Skeleton & Core API

Generic Test Harness

```
class Question {
    // — PROBLEM-SOLVING PROCESS —

    // 1. INPUT / OUTPUT : types, sizes, ranges;
    // what to return; examples.
    // 2. HIGH-LEVEL ALGO : pick the pattern (two-ptr
    // / BFS / DP / ...); state the idea in 1 line.
    // 3. IMPLEMENTATION : data structures,
    // loop/recursion, invariants.
    // 4. EDGE CASES : empty, size 0/1,
    // duplicates, negatives, overflow, all-same, not-found.
    // 5. COMPLEXITY : time & space; is it
    // within limits?
    // 6. OPTIMIZATION : reduce time/space
    // (hashing, two-pointer, rolling array, pruning, early
    // exit).
    //
}

private int data;
public Question(int data) {
    this.data = data;
}
public void solve() {
    // TODO
}

public class Solution {
    public static void main(String[] args) {
        Question q = new Question(0);
        q.solve();
    }
}
```

Naming Conventions (enforced across all template files)

Common Java API Cheat-Sheet

```
Map<Integer, Integer> map = new HashMap<>();
// key -> value
Map<Integer, List<Integer>> adj = new HashMap<>();
// adjacency / buckets
Set<Integer> set = new HashSet<>();
// membership
List<Integer> list = new ArrayList<>();
// dynamic array
Deque<Integer> stack = new ArrayDeque<>();
// STACK: push / pop / peek
Deque<Integer> queue = new ArrayDeque<>();
// QUEUE: offer / poll / peek

// (ArrayDeque beats java.util.Stack / LinkedList)
PriorityQueue<Integer> minHeap = new PriorityQueue<>();
// min-heap (default)
PriorityQueue<Integer> maxHeap = new PriorityQueue<>();
// max-heap
(Collections.reverseOrder());
PriorityQueue<int[]> pq = new PriorityQueue<>();
((a, b) -> a[1] - b[1]); // by field, min-first
TreeMap<Integer, Integer> tmap = new TreeMap<>();
// sorted keys: floor/ceiling/first/last
TreeSet<Integer> tset = new TreeSet<>();
// sorted set: floor/ceiling/higher/lower
LinkedHashSet<Integer> lset = new LinkedHashSet<>();
// insertion-ordered set (LFU buckets)
Random rand = new Random();
// rand.nextInt(n) -> 0..n-1 (RandomizedSet)
int[] arr = new int[n];
// defaults to 0
boolean[] seen = new boolean[n];
// defaults to false
int[][] grid = new int[m][n];
// 2D, all 0
int[] dr = {1,-1,0,0}, dc = {0,0,1,-1}; //
DOWN,UP,RIGHT,LEFT; iterate for(dir=0..3)
i+dr[dir],j+dc[dir]
```

```
// — Map —
map.put(k, v);
map.getOrDefault(k, 0);
// safe read with default
map.merge(k, 1, Integer::sum);
// ← counting frequency (most common)
map.computeIfAbsent(k, x -> new ArrayList<>()
    ).add(v); // ← build adjacency / buckets
map.putIfAbsent(k, v);
// ← write only first occurrence (keep earliest
    index)
map.containsKey(k); map.remove(k);
for (Map.Entry<Integer,Integer> e : map.entrySet()) {
    e.getKey(); e.getValue(); e.setValue(v); }
map.keySet(); map.values();
// — Set —
set.add(x); set.contains(x); set.remove(x);
// — List —
list.add(x); list.get(i); list.set(i, x);
list.remove(list.size() - 1);
// ← pop in backtracking
list.size(); list.isEmpty(); list.addAll(other);
Collections.sort(list);
// ascending
list.sort((a, b) -> b - a);
// custom / descending
Collections.reverse(list);
// ← reverse path after building it backwards
```

```
// Deque as STACK (LIFO) // Deque as
QUEUE (FIFO)
stack.push(x); queue.offer(x);
int top = stack.pop(); int head =
queue.poll();
int look = stack.peek(); int look =
queue.peek();
stack.isEmpty(); queue.isEmpty();
queue.isEmpty();
// Deque BOTH ENDS — needed for the MONOTONE DEQUE
(sliding-window max/min)
// and for addFirst-style level building (zigzag
BFS).
deque.offerLast(x); deque.pollLast();
deque.peekLast(); // back (push/pop/peek tail)
deque.offerFirst(x); deque.pollFirst();
deque.peekFirst(); // front (push/pop/peek head)
deque.addLast(x); deque.addFirst(x);
// same as offer*, throw if capacity-bound
// PriorityQueue (heap)
pq.offer(x); int best = pq.poll(); int look =
pq.peek(); pq.size();
pq.remove(x); // ← 0(n) removal of an arbitrary
element (lazy-deletion alternative in #480)
```

```
Integer.MAX_VALUE; // 2147483647 (2^31 - 1) -
sentinel for "min so far"
Integer.MIN_VALUE; // -2147483648 (-2^31) -
sentinel for "max so far"
Long.MAX_VALUE; // when sums can overflow int,
accumulate in long
int k = i + (j - i) / 2; // ← avoid (i +
j) overflow in binary search
Integer.compare(a, b); // ← overflow-
safe comparator; use instead of (a - b)
(a, b) -> Integer.compare(a[1], b[1]); // ← safe
PriorityQueue/Arrays.sort comparator
Double.compare(a, b); // ← comparator
for double[] heaps (probability, median)
```

```
ch - 'a'; // 'a'..'z' ->
0..25 (lowercase bucket index)
ch - '0'; // '0'..'9' ->
0..9 (digit value)
(char) ('a' + i); // 0..25 ->
'a'..'z'
num = num * 10 + (ch - '0'); // ← build a
number digit by digit
Integer.parseInt("123"); // String -> int
(throws on bad input)
Integer.valueOf(123); // int -> Integer
(boxed)
String.valueOf(123); // int/char/etc
-> String
Character.getNumericValue('7'); // char -> 7
// char classification
Character.isDigit(ch); Character.isLetter(ch);
Character.isLetterOrDigit(ch);
Character.isWhitespace(ch);
Character.toLowerCase(ch);
Character.toUpperCase(ch);
```

```
s.length(); s.charAt(i); s.substring(i, j);
// [i, j)
s.toCharArray(); s.split(" "); s.equals(t);
s.compareTo(t);
s.indexOf("ab"); s.isEmpty();
s.startsWith("ab"); s.endsWith("z");
s.contains("x"); // prefix / suffix / substring tests
s.toLowerCase(); s.toUpperCase();
// case-normalize (e.g. case-insensitive compare)
s.trim(); s.replace('a', 'b'); s.repeat(n);
// strip ends / swap chars / repeat
StringBuilder builder = new StringBuilder();
builder.append(x); builder.reverse();
builder.toString();
builder.charAt(i); builder.setCharAt(i, c);
builder.deleteCharAt(i); builder.length();
String.join(",", list);
// list -> "a,b,c"
```

```
Arrays.sort(arr);
// ascending in place
Arrays.sort(intervals, (a, b) -> a[0] - b[0]);
// 2D by first field
Arrays.fill(dp, Integer.MAX_VALUE);
// seed a dp/dist array
Arrays.equals(window, need);
// ← element-wise array compare (anagram check)
Arrays.copyOf(arr, len); Arrays.copyOfRange(arr, i,
j);
System.arraycopy(src, srcPos, dst, dstPos, len);
// ← fast block copy (merge step)
Arrays.asList(1, 2, 3);
// fixed-size List view
int sum = Arrays.stream(arr).sum();
int max = Arrays.stream(arr).max().getAsInt();
// List<Integer> -> int[]
int[] a =
list.stream().mapToInt(Integer::intValue).toArray();
List<Integer> l =
Arrays.stream(a).boxed().collect(Collectors.toList());
;
```

```
Math.max(a, b); Math.min(a, b); Math.abs(x);
Math.pow(a, b); Math.sqrt(x); Math.ceil(x);
Math.floor(x);
n & (n - 1); // clear lowest set bit
n & (-n); // isolate lowest set bit
1 << i; // bit i set (use 1L << i for
i ≥ 31)
(n >> i) & 1; // read bit i
(n & 1) == 0; // ← even (last bit 0)
(n & 1) == 1; // ← odd (last bit 1)
n >> 1; // ← divide by 2 (drop last
bit); n << 1 = multiply by 2
Integer.bitCount(n); // # of set bits
Integer.toBinaryString(n);
```

```
tmap.firstKey(); tmap.lastKey();
tmap.floorKey(x); // largest key ≤ x
tmap.ceilingKey(x); // smallest key ≥ x
tmap.lowerKey(x); // largest key < x
tmap.higherKey(x); // smallest key > x
tset.floor(x); tset.ceiling(x); tset.lower(x);
tset.higher(x); tset.first(); tset.last();
```

```
// — NEGATIVE-SAFE MODULO — replaces the manual ((x
% n) + n) % n
Math.floorMod(x, n); // always in
[0, n-1] even when x < 0 (prefix-sum % k)
// — int[] INSTEAD OF HashMap for a bounded key
space
int[] count = new int[26]; // 0(1) no-
boxing freq vs HashMap<Character,Integer>
count[ch - 'a']++; // also faster
to compare: Arrays.equals(a, b)
// — 1D ROLLING ARRAY for DP — drop a dimension
when row i only needs row i-1
int[] dp = new int[n + 1]; // 0(n) space
instead of 0(m·n)
for (...) for (int j = target; j >= w; j--) dp[j] +=
dp[j - w]; // 0/1 knapsack: iterate j DOWN
// — FIXED-SIZE HEAP for top-k — 0(n log k) beats
sorting 0(n log n)
if (heap.size() > k) heap.poll(); // keep only k
best; min-heap for "k largest"
// — LAZY DELETION over pq.remove(obj) — pq.remove
is 0(n); skip stale entries instead
if (entry[0] != dist[node]) continue; // Dijkstra:
ignore outdated heap entries
// — ArrayDeque over Stack / LinkedList — faster,
no synchronization, both ends 0(1)
Deque<Integer> stack = new ArrayDeque<>(); // never
use java.util.Stack
// — StringBuilder over String += in a loop — 0(n)
vs 0(n^2)
StringBuilder builder = new StringBuilder(); //
append, then builder.toString() once
// — EARLY EXIT / PRUNING — return the moment the
answer is decided
if (found) return result; // e.g.
backtracking bounds
```

Template Index

Two Pointers

Pattern 1 — Opposite Two Pointers

```
int i = 0, j = n - 1;
while (i < j) {
    if (condition == target) {
        /* record */ i++; j--;
    } else if (tooSmall) {
        i++;
    } else {
        j--;
    }
}
```

Sliding Window

The Three Templates

```
int i = 0;
for (int j = 0; j < n; j++) {
    // 1. add nums[j] to state
    if (j - i + 1 == k) {
        // 2. record (window is exactly size k)
        // 3. remove nums[i] from state
        i++;
    }
}

int i = 0, result = 0;
for (int j = 0; j < n; j++) {
    // 1. add nums[j] to state
    while (/* window exceeds constraint */) {
        i++;
    }
    result = Math.max(result, j - i + 1);
}

int i = 0, result = Integer.MAX_VALUE;
for (int j = 0; j < n; j++) {
    // 1. add nums[j] to state
    while (/* window VALID */) {
        result = Math.min(result, j - i + 1);
        i++;
    }
}
```

Frequency Map Trick (used by 567 and 76)

```
if (need[s.charAt(j)]-- > 0) required--;
if (need[s.charAt(i)]++ >= 0) required++;
i++;
```

Prefix Sum + HashMap

Core Template

```
Map<Long, Integer> map = new HashMap<>();
map.put(0L, ???);
long sum = 0;
for (int i = 0; i < nums.length; i++) {
    sum += nums[i];
    long lookup = transform(sum);
    map.put(storeKey(sum), storeValue(i)); // what to
    store (sum or sum%k → count or index)
}
```

Binary Search

The Only Template — [i, j], find first k with condition(k)

```
int i = 0, j = nums.length;
while (i < j) {
    int k = i + (j - i) / 2;
    if (!condition(k)) {
        i = k + 1;
    } else {
        j = k;
    }
}
```

How to Choose

Sorting

Template 1 — Merge Sort

```
public void mergeSort(int[] nums) {
    mergeSort(nums, 0, nums.length, new
    int[nums.length]);
}

private void mergeSort(int[] nums, int start, int
end, int[] temp) {
    if (end - start <= 1) return;
    int mid = start + (end - start) / 2;
    mergeSort(nums, start, mid, temp);
    mergeSort(nums, mid, end, temp);
    merge(nums, start, mid, end, temp);
}

private void merge(int[] nums, int start, int mid,
int end, int[] temp) {
    int i = start, j = mid, k = start;
    while (i < mid || j < end) {
        if (j >= end || (i < mid && nums[i] <=
        nums[j])) {
            temp[k++] = nums[i++];
        } else {
            temp[k++] = nums[j++];
        }
    }
    for (i = start; i < end; i++) nums[i] = temp[i];
}
```

Template 2 — Quick Sort (3-way Dutch Flag Partition)

```
public void quickSort(int[] nums) {
    quickSort(nums, 0, nums.length);
}

private void quickSort(int[] nums, int start, int
end) {
    if (end - start <= 1) return;
    int pivot = nums[start + (end - start) / 2];
    int i = start, k = start, j = end - 1;
    while (k <= j) {
        if (nums[k] < pivot) {
            swap(nums, k++, i++);
        } else if (nums[k] == pivot) {
            k++;
        } else {
            swap(nums, k, j--);
        }
    }
    quickSort(nums, start, i);
    quickSort(nums, k, end);
}

private void swap(int[] nums, int i, int j) {
    int t = nums[i]; nums[i] = nums[j]; nums[j] = t;
}
```

Template 3 — Quick Select (partial sort — k-th element)

```
private int quickSelect(int[] nums, int start, int
end, int target) {
    int pivot = nums[start + (end - start) / 2];
    int i = start, k = start, j = end - 1;
    while (k <= j) {
        if (nums[k] < pivot) {
            swap(nums, k++, i++);
        } else if (nums[k] == pivot) {
            k++;
        } else {
            swap(nums, k, j--);
        }
    }
    if (target < i) {
        return quickSelect(nums, start, i, target);
    } else if (target < k) {
        return pivot;
    } else {
        return quickSelect(nums, k, end, target);
    }
}
```

Template 4 — Counting Sort

```
public void countingSort(int[] arr) {
    if (arr.length == 0) return;
    int min = Arrays.stream(arr).min().getAsInt();
    int max = Arrays.stream(arr).max().getAsInt();
    int[] buckets = new int[max - min + 1];
    for (int x : arr) buckets[x - min]++;
    int i = 0;
    for (int v = 0; v < buckets.length; v++)
        while (buckets[v]-- > 0) arr[i++] = v + min;
}
```

Linked List

When to Use — Signal → Pattern All Four Templates

```
// 1. DELETE / FILTER - sentinel guards head
removal; previous tracks last kept node
ListNode sentinel = new ListNode(0);
sentinel.next = head;
ListNode previous = sentinel, current = head;
while (current != null) {
    if (shouldDelete(current)) {
        previous.next = current.next;
    } else {
        previous = current;
    }
    current = current.next;
}
return sentinel.next;

// 2. REVERSAL - re-wire one node at a time
ListNode previous = null, current = head;
while (current != null) {
    ListNode next = current.next;
    current.next = previous;
    previous = current;
    current = next;
}
return previous;

// 3. FAST / SLOW - two pointers at different speeds
ListNode slow = head, fast = head;
while (fast != null && fast.next != null) {
    slow = slow.next;
    fast = fast.next.next;
}

// 4. MERGE - sentinel collects nodes from two lists
in order
ListNode sentinel = new ListNode(0);
ListNode current = sentinel;
while (a != null && b != null) {
    if (a.val <= b.val) {
        current.next = a;
        a = a.next;
    } else {
        current.next = b;
        b = b.next;
    }
    current = current.next;
}
current.next = (a != null) ? a : b;
return sentinel.next;
```

String

Core Idioms

```
// 1. CHAR COUNT - lowercase letters
int[] count = new int[26];
for (char ch : s.toCharArray()) count[ch - 'a']++;
// 2. CHAR COUNT - digits 0-9
int[] count = new int[10];
for (char ch : s.toCharArray()) count[ch - '0']++;
// 3. CHAR TO INTEGER - build number digit by digit
int num = 0;
for (int i = 0; i < s.length(); i++)
    num = num * 10 + (s.charAt(i) - '0');
// 4. ANAGRAM HASH KEY - frequency-based (bucket sort
style) O(k) vs sort O(k log k)
String hashKey(String s) {
    int[] count = new int[26];
    for (char ch : s.toCharArray()) count[ch -
    'a']++;
    StringBuilder builder = new StringBuilder();
    for (int i = 0; i < 26; i++) {
        builder.append((char)('a' + i));
        builder.append(count[i]);
    }
    return builder.toString();
}

char[] chars = s.toCharArray();
Arrays.sort(chars);
String key = new String(chars);
// 5. BUCKET SORT - sort by frequency without
comparison sort
int[] count = new int[128];
for (char ch : s.toCharArray()) count[ch]++;
List<Character>[] buckets = new List[s.length() + 1];
// index = frequency
for (int i = 0; i < 128; i++) {
    if (count[i] > 0) {
        int freq = count[i];
        if (buckets[freq] == null) buckets[freq] =
        new ArrayList<>();
        buckets[freq].add((char) i);
    }
}

StringBuilder builder = new StringBuilder();
for (int freq = buckets.length - 1; freq > 0; freq--)
{
    if (buckets[freq] == null) continue;
    for (char ch : buckets[freq])
        for (int i = 0; i < freq; i++)
            builder.append(ch);
}

// 6. EXPAND FROM CENTER - palindrome check in O(1)
space
int expand(String s, int i, int j) {
    while (i >= 0 && j < s.length() && s.charAt(i) ==
    s.charAt(j)) { i--; j++; }
    return j - i - 1;
}
```

Stack / Queue / Heap

All Templates

```
// 1. STACK (LIFO) – use ArrayDeque, not Stack class
Deque<Integer> stack = new ArrayDeque<>();
stack.push(x);
stack.pop();
stack.peek();
// 2A. MONOTONE DECREASING STACK – next GREATER
element
Deque<Integer> stack = new ArrayDeque<>();
for (int i = 0; i < n; i++) {
    while (!stack.isEmpty() && nums[stack.peek()] <
        nums[i])
        result[stack.pop()] = nums[i];
    stack.push(i);
}
// 2B. MONOTONE INCREASING STACK – next SMALLER
element / span width
Deque<Integer> stack = new ArrayDeque<>();
for (int i = 0; i < n; i++) {
    while (!stack.isEmpty() && nums[stack.peek()] >
        nums[i]) {
        int height = nums[stack.pop()];
        int width = stack.isEmpty() ? i : i -
            stack.peek() - 1;
        stack.push(i);
    }
}
// 3. MONOTONE DEQUE – sliding window max/min
Deque<Integer> queue = new ArrayDeque<>();
for (int i = 0; i < n; i++) {
    while (!queue.isEmpty() && nums[queue.peekLast()]
        <= nums[i])
        queue.pollLast();
    queue.offerLast(i);
    if (queue.peekFirst() < i - k + 1)
        queue.pollFirst();
    if (i >= k - 1) result[i - k + 1] =
        nums[queue.peekFirst()];
}
// 4. PRIORITY QUEUE
PriorityQueue<Integer> minPQ = new PriorityQueue<>();
PriorityQueue<Integer> maxPQ = new PriorityQueue<>
    (Collections.reverseOrder()); // max-heap
PriorityQueue<int[]> pq = new PriorityQueue<>
    ((a, b) -> a[1] - b[1]); // custom: sort by index 1
pq.offer(x);
pq.poll();
pq.peek();
// 5. TWO HEAPS – maintain median in O(log n)
PriorityQueue<Integer> maxHeap = new PriorityQueue<>
    (Collections.reverseOrder());
PriorityQueue<Integer> minHeap = new PriorityQueue<>
    ();
void addNum(int num) {
    maxHeap.offer(num);
    minHeap.offer(maxHeap.poll());
    if (maxHeap.size() < minHeap.size())
        maxHeap.offer(minHeap.poll());
}
double findMedian() {
    return maxHeap.size() > minHeap.size()
        ? maxHeap.peek()
        : (maxHeap.peek() + minHeap.peek()) / 2.0;
}
```

BFS (Tree)

The Template — Level-Aware BFS

```
Queue<TreeNode> queue = new ArrayDeque<>();
queue.offer(root);
int level = 0;
while (!queue.isEmpty()) {
    int size = queue.size();
    level++;
    for (int i = 0; i < size; i++) {
        TreeNode node = queue.poll();
        if (node.left != null)
            queue.offer(node.left);
        if (node.right != null)
            queue.offer(node.right);
    }
}
```

When to Use — Signal → Pattern

The One Rule — Snapshot Level Size First

Always snapshot the level size BEFORE the inner **for** loop:

```
int size = queue.size();
for (int i = 0; i < size; i++) { ... }
```

Without the snapshot, newly added children mix with the current level
and $i == \text{size} - 1$ / $i == 0$ checks become meaningless.

DFS / Backtracking

When to Use — Signal → Pattern

All Templates

```
// 1. PROCESS CHILDREN FIRST, COMBINE AT NODE (post-order)
private int dfs(TreeNode node) {
    if (node == null) return BASE;
    int left = dfs(node.left);
    int right = dfs(node.right);
    return ...;
}
// 2. PROCESS NODE FIRST, PASS STATE DOWN (pre-order)
private void dfs(TreeNode node, int accumulated) {
    if (node == null) return;
    accumulated += node.val;
    if (isLeaf(node)) { /* record */ return; }
    dfs(node.left, accumulated);
    dfs(node.right, accumulated);
}
// 3. TREE – GLOBAL ANSWER (return up the best single arm; update global across both arms)
int answer = Integer.MIN_VALUE;
private int dfs(TreeNode node) {
    if (node == null) return 0;
    int left = Math.max(0, dfs(node.left));
    int right = Math.max(0, dfs(node.right));
    answer = Math.max(answer, left + right +
        node.val);
    return Math.max(left, right) + node.val;
}
// 4. GRID DFS – mark visited by mutating grid; 4-
directional flood fill
private void dfs(int[][] grid, int r, int c) {
    if (r < 0 || r >= grid.length || c < 0 || c >=
        grid[0].length) return;
    if (grid[r][c] != TARGET) return;
    grid[r][c] = VISITED;
    dfs(grid, r+1, c); dfs(grid, r-1, c);
    dfs(grid, r, c+1); dfs(grid, r, c-1);
}
// 5. GRAPH DFS – 3-color visited: 0=unseen 1=in-
stack 2=done
int[] state;
private boolean dfs(int node) {
    if (state[node] == 1) return true;
    if (state[node] == 2) return false;
    state[node] = 1;
    for (int neighbor : graph.get(node))
        if (dfs(neighbor)) return true;
    state[node] = 2;
    return false;
}
// 6. BACKTRACKING – choose / recurse / undo
private void backtrack(int start, List<Integer>
    current) {
    if (baseCase) { result.add(new ArrayList<>
        (current)); return; }
    for (int i = start; i < n; i++) {
        if (skipCondition(i)) continue;
        current.add(nums[i]);
        backtrack(next, current);
        current.remove(current.size()-1);
    }
}
```

Graph

Complexity Legend

BFS / DFS	$O(V + E)$	visit each
vertex and edge once		
Topo sort (Kahn)	$O(V + E)$	each node
enqueued once, each edge relaxed once		
Union-Find	$\sim O(n \cdot \alpha(n)) \approx O(n)$	α = inverse
Ackermann, effectively constant		
Dijkstra	$O((V + E) \log V)$	non-negative
weights ONLY		

Graph Representation

```
List<List<Integer>> graph = new ArrayList<>();
for (int i = 0; i < n; i++) graph.add(new ArrayList<>
    ());
for (int[] edge : edges) {
    graph.get(edge[0]).add(edge[1]);
    graph.get(edge[1]).add(edge[0]);
}
List<List<int[]>> graph = new ArrayList<>();
for (int i = 0; i < n; i++) graph.add(new ArrayList<>
    ());
for (int[] edge : edges)
    graph.get(edge[0]).add(new int[] {edge[1],
        edge[2]});
int[] dr = {1, -1, 0, 0};
int[] dc = {0, 0, 1, -1};
```

Core Templates

```
// 1. BFS – shortest path / level order (track
distance by level-size snapshot)
Queue<Integer> queue = new ArrayDeque<>();
boolean[] visited = new boolean[n];
queue.offer(start);
visited[start] = true;
int level = 0;
while (!queue.isEmpty()) {
    int size = queue.size();
    for (int i = 0; i < size; i++) {
        int node = queue.poll();
        for (int neighbor : graph.get(node)) {
            if (!visited[neighbor]) {
                visited[neighbor] = true;
                queue.offer(neighbor);
            }
        }
    }
    level++;
}
// 2. TOPOLOGICAL SORT – Kahn's BFS
int[] inDegree = new int[n];
for (int[] edge : edges) inDegree[edge[1]]++;
Queue<Integer> queue = new ArrayDeque<>();
for (int i = 0; i < n; i++) if (inDegree[i] == 0)
    queue.offer(i);
List<Integer> order = new ArrayList<>();
while (!queue.isEmpty()) {
    int node = queue.poll();
    order.add(node);
    for (int neighbor : graph.get(node))
        if (--inDegree[neighbor] == 0)
            queue.offer(neighbor);
}
// 3. UNION FIND
int[] parent, rank;
void init(int n) {
    parent = new int[n]; rank = new int[n];
    for (int i = 0; i < n; i++) parent[i] = i;
}
int find(int x) {
    if (parent[x] != x) parent[x] = find(parent[x]);
    return parent[x];
}
boolean union(int x, int y) {
    int px = find(x), py = find(y);
    if (px == py) return false;
    if (rank[px] < rank[py]) { int t = px; px = py;
        py = t; }
    parent[py] = px;
    if (rank[px] == rank[py]) rank[px]++;
    return true;
}
// 4. DIJKSTRA – shortest path (non-negative weights)
int[] dist = new int[n];
Arrays.fill(dist, Integer.MAX_VALUE);
dist[src] = 0;
PriorityQueue<int[]> pq = new PriorityQueue<>((a, b)
    -> a[1] - b[1]);
pq.offer(new int[] {src, 0});
while (!pq.isEmpty()) {
    int[] current = pq.poll();
    int node = current[0], d = current[1];
    if (d > dist[node]) continue;
    for (int[] nb : graph.get(node)) {
        int next = nb[0], w = nb[1];
        if (dist[node] + w < dist[next]) {
            dist[next] = dist[node] + w;
            pq.offer(new int[] {next, dist[next]});
        }
    }
}
// 5. BIPARTITE CHECK – BFS 2-coloring
int[] color = new int[n];
Arrays.fill(color, -1);
Queue<Integer> queue = new ArrayDeque<>();
for (int start = 0; start < n; start++) {
    if (color[start] != -1) continue;
    color[start] = 0;
    queue.offer(start);
    while (!queue.isEmpty()) {
        int node = queue.poll();
        for (int neighbor : graph.get(node)) {
            if (color[neighbor] == -1) {
                color[neighbor] = 1 - color[node];
                queue.offer(neighbor);
            } else if (color[neighbor] == color[node])
                return false;
        }
    }
}
```

```
}
return true;

Greedy

Two Interval Templates

MERGE DIRECTION RULE (for merge-style problems like #56, where both keys work):
    Sort by START → traverse FORWARD (i = 0 → n-1), extend the END of last kept
    Sort by END → traverse BACKWARD (i = n-1 → 0), extend the START of last kept
    These two are exact mirror images. See #56 for both written out side by side.
    (NOTE: this mirror only applies to merging. Template 1 below sorts by end but still sweeps FORWARD – there the goal is counting, not merging.)
    TEMPLATE 1 – Sort by END time, sweep forward, track end of last kept interval
        Use when: maximizing count of non-overlapping intervals
        Greedy insight: earliest-finishing interval leaves most room for future ones
    TEMPLATE 2 – Sort by START time, sweep forward, merge when overlapping
        Use when: merging/counting overlapping intervals
        Greedy insight: processing in start order → each interval only needs to compare with the last merged result
```

```
Arrays.sort(intervals, (a, b) -> a[1] - b[1]);
int lastEnd = Integer.MIN_VALUE;
for (int[] interval : intervals) {
    if (interval[0] >= lastEnd) {
        lastEnd = interval[1];
    } else {
    }
}
Arrays.sort(intervals, (a, b) -> a[0] - b[0]);
List<int[]> result = new ArrayList<>();
for (int[] interval : intervals) {
    if (result.isEmpty() ||
    result.get(result.size()-1)[1] < interval[0]) {
        result.add(interval);
    } else {
        result.get(result.size()-1)[1] =
            Math.max(result.get(result.size()-1)[1],
            interval[1]);
    }
}
Arrays.sort(intervals, (a, b) -> a[0] - b[0]);
PriorityQueue<Integer> pq = new PriorityQueue<>();
for (int[] interval : intervals) {
    if (!pq.isEmpty() && pq.peek() <= interval[0])
        pq.poll();
    pq.offer(interval[1]);
}
```

Dynamic Programming

All Six Templates

```
// 1. LINEAR 1D – each cell depends on a fixed number of previous cells
int[] dp = new int[n + 1];
dp[0] = base;
for (int i = 1; i <= n; i++)
    dp[i] = f(dp[i-1], dp[i-2], ...);
// KADANE (max subarray, #53) = linear-1D where dp[i] = best sum ENDING at i:
// 2A. LOOK-BACK 1D – dp[i] depends on all j < i
int[] dp = new int[n];
for (int i = 0; i < n; i++) {
    for (int j = 0; j < i; j++) {
        dp[i] = f(dp[i], dp[j]);
    }
}
// 3. 2D SEQUENCE – match consumes both, mismatch drops one side. WHEN: LCS / edit
// 4. INTERVAL DP – short ranges first (i right-to-left, j from i+1; dp[i+1][*] ready).
// 5. GRID DP – each cell accumulates from cells it could arrive from (up/left).
// 6. STATE MACHINE – named states updated from yesterday's states each step.
```

```
Want count or min/max?
├── Count (dp[j] += dp[j - num])
│   └── 0/1 (each item once): j descending → #416,
#494
├── Unbounded (reuse): j ascending → #518
│   └── Ordered (perms): target outer, nums
inner → #377
├── Min/Max (dp[j] = min/max(...))
│   └── 0/1 minimize: j descending → #474
(maximize)
└── Unbounded minimize: j ascending → #322
```

```
// — 0/1 KNAPSACK — each item at most once
int[] dp = new int[target + 1];
dp[0] = 1;
for (int i = 0; i < nums.length; i++) {
    for (int j = target; j >= nums[i]; j--)
        dp[j] += dp[j - nums[i]];
}
// — UNBOUNDED KNAPSACK — each item reusable
int[] dp = new int[target + 1];
dp[0] = 1;
for (int i = 0; i < nums.length; i++) {
    for (int j = nums[i]; j <= target; j++)
        dp[j] += dp[j - nums[i]];
}
// — PERMUTATION COUNT — order matters (Combination Sum IV)
int[] dp = new int[target + 1];
dp[0] = 1;
for (int j = 1; j <= target; j++) {
    for (int num : nums) {
        if (j >= num) dp[j] += dp[j - num];
    }
}
}
```

```
int[][] dp = new int[m + 1][n + 1];
for (int i = 1; i <= m; i++) {
    for (int j = 1; j <= n; j++) {
        if (s1.charAt(i-1) == s2.charAt(j-1)) {
            dp[i][j] = dp[i-1][j-1] + 1;
        } else {
            dp[i][j] = combine(dp[i-1][j], dp[i][j-1], dp[i-1][j-1]);
        }
    }
}
```

```
int[][] dp = new int[n][n];
for (int i = 0; i < n; i++) dp[i][i] = base_case;
for (int i = n - 1; i >= 0; i--) {
    for (int j = i + 1; j < n; j++) {
        dp[i][j] = ...;
    }
}
```

```
int[][] dp = new int[n][n];
for (int i = 0; i < n; i++) dp[i][i] = base_case;
for (int i = 0; i < n; i++) {
    for (int j = i - 1; j >= 0; j--) {
        dp[i][j] = ...;
    }
}
```

```
int[][] dp = new int[n][n];
for (int i = 0; i < n; i++) dp[i][0] = 1;
for (int i = 1; i < n; i++) {
    for (int j = 1; j <= i; j++) {
        dp[i][j] = ...;
    }
}
```

```
int[] dr = {1,-1,0,0}, dc = {0,0,1,-1};
int[][] dp = new int[rows][cols];
for (int i = 0; i < rows; i++)
    for (int j = 0; j < cols; j++)
        dp[i][j] = grid[i][j] + f(dp[i-1][j], dp[i][j-1]);
int[][] memo = new int[rows][cols];
for (int i = 0; i < rows; i++)
    for (int j = 0; j < cols; j++)
        dfs(grid, i, j, memo, dr, dc);
```

```
cash = max profit when NOT holding (free to buy or idle)
hold = max profit when HOLDING stock (bought it at some cost)
cooldown = max profit right after selling (can't buy today)
```

	buy	sell
re-buy condition	hold = max(hold, -price)	
#121 (1 tx):	no re-buy (ignore cash)	
#122 (unlimited):	hold = max(hold, cash - price)	
re-buy with full cash		
#123 (2 tx):	chain: buy2 uses sell1 cash	
4 states chained		
#309 (cooldown):	hold = max(hold, cooldown - price)	
must wait 1 day		
#714 (fee):	hold = max(hold, cash - price)	
pay fee on sell		

Bit Manipulation

```
Canonical Template

// — XOR CANCEL — pairs cancel; lone element survives
int result = 0;
for (int num : nums) result ^= num;
// — BIT OPERATIONS —
boolean isSet = ((n >> i) & 1) == 1;
int set = n | (1 << i);
int clear = n & ~(1 << i);
int toggle = n ^ (1 << i);
int lowest = n & (-n);
boolean isPow2 = n > 0 && (n & (n-1)) == 0;
// — COUNT SET BITS — Brian Kernighan: each iteration removes one set bit
int count = 0;
while (n != 0) { count++; n &= (n - 1); }
// — BITMASK AS SET — 26-bit integer represents presence of 26 letters
int mask = 0;
for (char c : word.toCharArray()) mask |= (1 << (c - 'a'));
if ((mask[i] & mask[j]) == 0) { /* no shared letter */ }
```

Variations

```
int ones = 0, twos = 0;
for (int num : nums) {
    ones = (ones ^ num) & ~twos;
    twos = (twos ^ num) & ~ones;
}
int xor = 0;
for (int num : nums) xor ^= num;
int diff = xor & (-xor);
int a = 0;
for (int num : nums)
    if ((num & diff) != 0) a ^= num;
int b = xor ^ a;
int[] dp = new int[n + 1];
for (int i = 1; i <= n; i++)
    dp[i] = dp[i >> 1] + (i & 1);
int shift = 0;
while (left != right) { left >>= 1; right >>= 1; shift++; }
return left << shift;
int add(int a, int b) {
    while (b != 0) {
        int carry = a & b;
        a = a ^ b;
        b = carry << 1;
    }
    return a;
}
```

Design

Canonical Template — HashMap + Doubly Linked List (LRU)

```
class CacheNode {
    int key, value;
    CacheNode previous, next;
    CacheNode(int key, int value) { this.key = key; this.value = value; }
}
// — SENTINEL NODES — head (MRU side) ↔ ... ↔ tail (LRU side)
CacheNode head = new CacheNode(0, 0), tail = new CacheNode(0, 0);
head.next = tail; tail.previous = head;
// — REMOVE NODE
void remove(CacheNode node) {
    node.previous.next = node.next;
    node.next.previous = node.previous;
}
// — INSERT AFTER HEAD (= mark as MRU) —
void insertAfterHead(CacheNode node) {
    node.next = head.next;
    node.previous = head;
    head.next.previous = node;
    head.next = node;
}
// — LRU EVICTION — tail.previous is LRU
CacheNode lru = tail.previous;
remove(lru);
map.remove(lru.key);
```

Variations

Math

Canonical Templates

```
// — DIGIT EXTRACTION LOOP — peel digits right-to-left
while (n != 0) {
    int digit = n % 10;
    n /= 10;
}
// — FAST POWER (exponentiation by squaring) —
double fastPow(double x, long n) {
    double result = 1;
    while (n > 0) {
        if ((n & 1) == 1) result *= x;
        x *= x;
        n >>= 1;
    }
    return result;
}
// — BASE CONVERSION — generic base b, 0-indexed
StringBuilder builder = new StringBuilder();
while (num > 0) {
    builder.append(num % b);
    num /= b;
}
builder.reverse();
int value = 0;
for (int digit : digits) value = value * b + digit;
// — SIEVE OF ERATOSTHENES — mark composites up to n
boolean[] composite = new boolean[n];
for (int i = 2; i * i < n; i++) {
    if (composite[i]) continue;
    for (int j = i * i; j < n; j += i)
        composite[j] = true;
}
```